

Retained Moments and Discovered Neural Directions

Exact Failure Diagnostics and Online Learning Experiments

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Abstract

What information lets a learner discover useful structural changes that were not individually specified before the data? We investigate this question by first attempting to reduce its natural formulations to existing results. Generic retained-moment characterization, adaptive use of a uniformly valid coresets, and residual spectral growth fail as substantial originality claims. We then study an exact obstruction in centered quadratic neural learning: subtracting the current model from a stored estimate of a population moment is different from recomputing empirical residual correlations. Without refitting, repeated spectral births from a fixed moment are exactly one spectral truncation. At a noiseless exact fit, the former construction almost surely proposes a harmful unthresholded birth, whereas sample residuals vanish. We derive its exact finite-sample variance, a fresh-residual contraction, and a conditional width–risk bound, with complete proofs and explicit classical dependencies. Reproducible estimator diagnostics and trained, ordinary-start online neural comparisons distinguish supplied directions, sample-discovered directions, refitting, retention, and fixed capacity. Independent reviews and negative controls are retained. These results provide an auditable research record and falsification tools; they do not establish a substantial new general theorem of structural learning, a universally better learner, or publication readiness.

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1 Question, evidence, and scientific status

The motivation is information-driven adaptation of neural structure. Neither entropy, neuron splitting, nor memory is imposed as the answer. The desired contribution must explain a consequence not already supplied by the closest external theorem or algorithm. This report records actual progress toward that standard, including candidates that did not meet it.

The initial question concerned information a learner should retain to *discover* future useful operations, rather than merely validate a supplied one. Discovery requires an optimization procedure that returns trainable weights, and its search work and information access must be counted. A theorem that estimates the supremum of possible gains does not necessarily return a maximizing neuron. Conversely, when a guarantee is uniform over a fixed function class, the eventual candidate identities may be chosen after the data without creating an additional validity problem.

The actual starting workspace matters. It already contains finite-encoder obstructions; a restricted activation-observation separation and high-order splits; repeated neural growth with fresh audits; exact retained-moment ambiguity and residual testing costs; and a 336-trajectory audit-release study. The latter finds benefits from releasing old labels, no resolved advantage of residual over random proposals, and stronger fixed online controls on some tasks. None is claimed again here. The starting 1,087 non-environment files were hashed and preserved. The requested Phase IV brief and handoff were read in full; the later actual artifacts superseded incomplete descriptions in the handoff.

Independent investigations considered retained decision information, noise-aware rank adaptation, coordinated high-order births, and function-preserving changes to trainability. Three reductions explain why several appealing central claims were rejected. The subsequent analysis changes the scientific target to a concrete finite-sample discrepancy: *what does a stored population inverse fail to subtract when the model has already learned part of the signal?* This has an exact answer in a realizable quadratic neural family. Its interpretation is deliberately bounded: the answer specializes matrix regression and stochastic approximation. Correctness and usefulness do not establish substantial external novelty.

2 The novelty checkpoint and nearest results

2.1 Reductions that defeated initial candidate claims

For unknown label means g and retained linear observations Λg , a best-action value of the form $\sup_{\theta} \langle h_{\theta}, g \rangle$ is a supremum of linear functionals. Foucart’s nonlinear optimal-recovery framework covers this object and bounded observation error [7, Theorems 1, 3; Section 2.3]. It does not

solve the neural maximizer problem. Moving from one supplied action to a continuum therefore does not by itself establish a new recovery theorem.

Suppose a retained objective obeys $\sup_{f \in \mathcal{F}} |F(f) - \tilde{F}(f)| \leq \varepsilon$ on one event. Then all adaptively chosen $f \in \mathcal{F}$ are covered on that event. If $\tilde{F}(\hat{f}) \leq \inf_{f \in \mathcal{F}} \tilde{F}(f) + \eta$, then

$$F(\hat{f}) \leq \inf_{f \in \mathcal{F}} F(f) + 2\varepsilon + \eta.$$

This follows by inserting the two approximation inequalities around the retained-objective optimization inequality. Strong coresets already have this simultaneous-query meaning [3]. The construction size, changing encompassing class, and optimization error remain substantive; renaming the final query a structural action does not change the argument.

Similarly, uniform score error ε and optimization error η produce a feature oracle with error $2\varepsilon + \eta$. Residual feature search and approximate conditional-gradient oracles are existing tools [1, Section 2.5]. Gradient-matching retention already targets models away from the current fit [2]; its Appendix A.4 discusses the loss of exact historical accumulation when embeddings change. Gradient mismatch also already enters convergence guarantees [11]. These sources prevent claiming that a replay buffer or retained probe gradients constitute the main contribution.

2.2 Why quadratic neurons do not make rank learning new

A signed quadratic network has matrix parameter $A = \sum_j a_j w_j w_j^\top$. Thus adding a neuron is adding a symmetric rank-one atom. GECCO already chooses a leading singular direction and refits coefficients [14]; under directional smoothness it supplies an $O(\|A\|_*^2/k)$ approximation comparison. Restricted-curvature greedy analysis yields stronger rates [10, Theorem 4]. The nonlinear factor optimization used in our experiments is not guaranteed to perform GECCO's exact corrective minimization.

Rank-one quadratic measurements already admit stable low-rank recovery [5, Theorem 1]. Fixed overparameterized quadratic networks also learn directions [12, Theorem 1.2], and incremental-rank trajectories of gradient descent are established under matrix-sensing assumptions [9, Theorem 4.1]. Raw Gaussian outer-product sensing does not satisfy ordinary Frobenius ℓ_2/ℓ_2 RIP merely because the covariates are Gaussian. This blocks an otherwise tempting incorrect transfer.

The residual matrix update analyzed below is stochastic gradient descent in matrix coordinates. With rank projection it has the stochastic iterative hard-thresholding form [13, Algorithm 1, Theorem 1]. Our exact Gaussian moment calculation directly handles fresh unbounded Gaussian batches, but does not supply a novel optimization principle. Signal-induced failure of untrimmed spectral initialization also has explicit prior analysis [4, Section 2.2]. Empirical residual cross-moments and spectral neural additions have a close structural-learning precedent [15, Propositions 3.2–3.3].

2.3 A theorem-level comparison summary

Claim	Closest result / mapping	Disposition and missing step
Best future gain from moments	Sup-linear optimal recovery [7]	Inherited value recovery; finding a trainable maximizer within a resource budget is additional.
Adaptive retained objective	Strong coresets [3]	Uniformity already covers data-dependent identities in its fixed class; changing-class online construction is not supplied.

Claim	Closest result / mapping	Disposition and missing step
Residual neuron discovery	Convex neural oracle and GECO [1, 14]	Rank-one atom search is known; no new direction-search theorem here.
Fixed- summary birth path	Spectral decomposition	Exact diagnostic; with no refit the path equals a single hard truncation.
Fresh residual correction	Stochastic gradient / StoIHT [13]	Exact Gaussian constants and a noise-free false-growth witness are proved; $O(d^2)$ batch bound is not an optimal low-rank rate.
Selected width certificate	Hard threshold / penalized rank	Conditional perturbation and oracle inequality; the stated finite-sample radius is conservative and requires norm bounds.
Actual trained histories	Present saved runs	Empirical evidence specific to observed-data proposals, optimizer, tasks and accounting; no theorem of universal structural superiority.

The full `NOVELTY_MATRIX.md` identifies source versions, observation interfaces, resources, guarantees, attempted reductions, and unresolved risks. The current representation-certification source was checked at v2, not inherited as v1 [8]; its repair discussion is outside the fixed kernel result. A source-level counterexample warning about a different current spectral claim is included in Appendix A; we do not use that claim as a black-box theorem.

3 An exact quadratic diagnostic and its limits

The results in this section are specializations of classical Gaussian moment identities, spectral approximation, and stochastic approximation. Their purpose is to distinguish population inversion, finite-sample estimation, and actual empirical optimization. They do not establish a substantially original learning algorithm or an advantage of autonomous neural growth. All probability and expectation statements below are exact under the stated model; the numerical checks in the experiments are separate evidence.

Fix integers $d, n \geq 1$. Matrices are real and symmetric unless indicated otherwise, with inner product $\langle A, B \rangle = \text{tr}(AB)$. Write $\|\cdot\|_F$, $\|\cdot\|_{\text{op}}$, and $\|\cdot\|_*$ for Frobenius, operator, and nuclear norms. Set

$$X \sim N(0, I_d), \quad H(X) = XX^\top - I_d, \quad q_A(X) = \langle A, H(X) \rangle.$$

The unknown fixed teacher is A_* . Labels obey

$$Y = q_{A_*}(X) + \xi, \quad \mathbb{E}\xi = 0, \quad \mathbb{E}\xi^2 = \sigma^2 < \infty, \quad \xi \text{ independent of } X. \quad (1)$$

Distinct examples have independent copies of (X, ξ) . Define population squared risk $R(B) = \mathbb{E}[(Y - q_B(X))^2]$.

A signed centered quadratic network

$$f(x) = \sum_{j=1}^k a_j \{(w_j^\top x)^2 - \|w_j\|_2^2\}$$

represents $B = \sum_{j=1}^k a_j w_j w_j^\top$. Its minimum width is $\text{rank}(B)$: any such sum has rank at most k , while an eigendecomposition represents B with exactly its number of nonzero eigenvalues.

The zero predictor has width zero. A constant output can implement the centering. Negative a_j are allowed throughout; restricting all outputs to be nonnegative changes the operation class to positive semidefinite matrices and does not preserve every result below.

3.1 Population geometry

Theorem 3.1 (T1: Gaussian quadratic geometry). *For symmetric A, B ,*

$$\mathbb{E}[q_A(X)q_B(X)] = 2 \operatorname{tr}(AB). \quad (2)$$

Consequently, under (1),

$$R(B) - R(A_*) = 2\|B - A_*\|_F^2, \quad \frac{1}{2}\mathbb{E}[YH(X)] = A_*. \quad (3)$$

Proof. For standard Gaussian coordinates, second and fourth moments are

$$\mathbb{E}[X_i X_j] = \delta_{ij}, \quad \mathbb{E}[X_i X_j X_k X_l] = \delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}.$$

Subtracting the product of second moments yields $\mathbb{E}[H_{ij}H_{kl}] = \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}$. Multiply by $A_{ij}B_{kl}$ and sum all indices. The symmetry of A, B gives (2). Taking one matrix to be an arbitrary symmetric test matrix gives $\mathbb{E}[q_A(X)H(X)] = 2A$.

Let $E = A_* - B$. Independence and zero mean of ξ give $\mathbb{E}[q_E(X)] = 0$ and $\mathbb{E}[\xi H(X)] = 0$. Thus $R(B) = \mathbb{E}[q_E(X)^2] + \sigma^2 = 2\|E\|_F^2 + \sigma^2$, while $R(A_*) = \sigma^2$. The second identity follows by taking the moment of Y . $\square$

Without centering, the corresponding quadratic prediction difference has second moment $2\|A - B\|_F^2 + \operatorname{tr}(A - B)^2$. Thus centering is part of the model, not an inconsequential convention.

3.2 What repeated growth from a fixed summary computes

Theorem 3.2 (T2: fixed-summary spectral growth). *Fix a symmetric matrix S and set $B_0 = 0$. At each nonterminal step, select a unit eigenvector v_t of $S - B_t$ whose eigenvalue λ_t has maximum absolute value, and set*

$$B_{t+1} = B_t + \lambda_t v_t v_t^\top.$$

For a fixed $\tau \geq 0$, terminate before this update if $|\lambda_t| \leq \tau$. Every nonzero update selects a previously unselected eigencomponent of S , in nonincreasing absolute-eigenvalue order. In at most d updates the terminal result is

$$B_{\text{stop}} = \sum_{j: |s_j| > \tau} s_j u_j u_j^\top, \quad S = \sum_{j=1}^d s_j u_j u_j^\top. \quad (4)$$

Intermediate truncations can be nonunique at ties. The final strict-threshold result is invariant to these choices.

Proof. We maintain an orthonormal set of selected eigenvectors of S such that B_t is the sum of their original eigencomponents. This holds at $t = 0$. The residual $S - B_t$ is zero on their span and equals S on its orthogonal complement. If a nonzero residual eigenvalue remains, every unit eigenvector with that nonzero eigenvalue is orthogonal to the selected span. It is an eigenvector of S with the same eigenvalue. The update therefore extends the orthonormal selected set and removes exactly that one component from the residual. Selecting maximum absolute value establishes the order. At most d nonzero components can be removed. The stopping rule removes all and only components with absolute eigenvalue strictly greater than τ , proving (4).

Within an eigenspace with the same signed eigenvalue, any orthonormal basis is permissible. If s and $-s$ are tied in absolute value, one may choose either eigenspace, but arbitrary mixtures across them need not be eigenvectors and are not allowed by the algorithm. When the residual is zero, the rule terminates, including when $\tau = 0$. $\square$

The directions in this construction can be learned from labels through S . Nevertheless, after S has been formed the trajectory is precisely an eigendecomposition procedure. A one-shot estimator with the same S , threshold and eigensolver produces the same predictor. This statement does not identify the trajectory of a method that subsequently refits neurons using empirical loss.

For an observed dataset, define

$$S_n = \frac{1}{2n} \sum_{i=1}^n Y_i H_i, \quad T_n(B) = \frac{1}{2n} \sum_{i=1}^n \langle B, H_i \rangle H_i, \quad H_i = H(X_i). \quad (5)$$

The actual residual moment satisfies the algebraic identity

$$G_n(B) := \frac{1}{2n} \sum_{i=1}^n \{Y_i - q_B(X_i)\} H_i = S_n - T_n(B). \quad (6)$$

It is the negative Euclidean matrix gradient of $L_n(B) = (4n)^{-1} \sum_i \{Y_i - q_B(X_i)\}^2$. The surrogate $S_n - B$ replaces the empirical operator T_n by its population mean. For fixed B independent of the sample, $\mathbb{E}[T_n(B) \mid B] = B$; equality on the realized sample does not follow. If B uses the same sample, even this conditional identity can fail.

Remark 3.3 (An explicit adaptive-reuse counterexample). Take $d = n = 1$, $A_* = 1$, $\sigma = 0$, and $H = X^2 - 1$. Then $Y = H$. Choose $B = S_1 = H^2/2$, which is a legitimate data-dependent estimate. Equation (5) gives $T_1(B) = B^2$. Thus $\mathbb{E}[T_1(B) \mid B] = B^2$, generally different from B . A same-sample residual correction is $C = B + G_1(B) = 2B - B^2$, so $|C - 1|^2 = (B - 1)^4$. The fresh-block conditional variance in the next theorem would instead be $14(B - 1)^2$ and does not apply here.

For exact empirical residual evaluation at arbitrary B , one must apply T_n , using replay, its stored operator, or an equivalent representation. In orthonormal symmetric coordinates, T_n is one half the empirical Gram operator of the $d(d+1)/2$ centered quadratic features. Its entries include fourth-order input moments. This is the ordinary regression normal equation; it is not a lower bound on physical memory, and no minimum-bit claim follows from it.

3.3 Exact sampling variance and fresh corrections

Theorem 3.4 (T3: variance of a fresh residual correction). *Define*

$$c_d = \frac{d^2 + 9d + 14}{2}.$$

For any fixed symmetric E and $Z_E = \frac{1}{2} q_E(X) H(X)$,

$$\mathbb{E} Z_E = E, \quad \mathbb{E} \|Z_E - E\|_F^2 = c_d \|E\|_F^2 + 2 \operatorname{tr}(E)^2. \quad (7)$$

Let $\mathcal{F}$ contain the complete history before a new block, and let B be a square-integrable symmetric $\mathcal{F}$ -measurable matrix. Conditional on $\mathcal{F}$, the new n observations are independent copies of (1) and independent of $\mathcal{F}$. Set

$$E = A_* - B, \quad C = B + \frac{1}{2n} \sum_{i=1}^n \{Y_i - q_B(X_i)\} H_i.$$

Then

$$\mathbb{E}[C \mid \mathcal{F}] = A_*, \quad (8)$$

$$\mathbb{E}[\|C - A_*\|_F^2 \mid \mathcal{F}] = \frac{c_d \|E\|_F^2 + 2 \operatorname{tr}(E)^2 + \sigma^2 d(d+1)/4}{n}. \quad (9)$$

Proof. Write $s = \|X\|_2^2$ and $U = X/\|X\|_2$, with any definition at $X = 0$, a null event. The radius $s \sim \chi_d^2$ is independent of the uniform direction U on the unit sphere. Put $\theta = \text{tr}(E)$ and $v = \|E\|_F^2$. Rotational symmetry, or the Gaussian fourth-moment identity applied before dividing out the radius, gives

$$\mathbb{E}[U^\top EU] = \frac{\theta}{d}, \quad \mathbb{E}[(U^\top EU)^2] = \frac{\theta^2 + 2v}{d(d+2)}.$$

Indeed, the second formula follows from $\mathbb{E}[(X^\top EX)^2] = \theta^2 + 2v$ and $\mathbb{E}[s^2] = d(d+2)$ by radial independence. Since $q_E = sU^\top EU - \theta$,

$$\mathbb{E}[q_E^2 | s] = \frac{(\theta^2 + 2v)s^2}{d(d+2)} - \frac{2\theta^2 s}{d} + \theta^2. \quad (10)$$

The chi-square moments are

$$\mathbb{E}[s^j] = \prod_{l=0}^{j-1} (d+2l), \quad j = 1, 2, 3, 4.$$

Substitute these into (10), after multiplying respectively by 1, s , s^2 , to obtain

$$\begin{aligned} \mathbb{E}[q_E^2] &= 2v, \\ \mathbb{E}[q_E^2 s] &= 2(d+4)v, \\ \mathbb{E}[q_E^2 s^2] &= 8\theta^2 + 2(d+4)(d+6)v. \end{aligned}$$

Also $\|H\|_F^2 = s^2 - 2s + d$. Hence

$$\begin{aligned} \mathbb{E}[q_E^2 \|H\|_F^2] &= 8\theta^2 + 2\{(d+4)(d+6) - 2(d+4) + d\}v \\ &= 8\theta^2 + 2(d^2 + 9d + 16)v. \end{aligned}$$

By Theorem 3.1, $\mathbb{E}Z_E = E$. The identity $\mathbb{E}\|Z_E - E\|_F^2 = \mathbb{E}\|Z_E\|_F^2 - \|E\|_F^2$ now proves (7).

For the noisy single-example correction, let $W = Z_E + \frac{1}{2}\xi H$. Conditional on the past, E is fixed. Independence and zero mean of ξ eliminate all cross terms between $Z_E - E$ and ξH . Furthermore,

$$\mathbb{E}\|H\|_F^2 = d(d+2) - 2d + d = d(d+1).$$

Consequently $\mathbb{E}[W | \mathcal{F}] = E$ and

$$\mathbb{E}[\|W - E\|_F^2 | \mathcal{F}] = c_d \|E\|_F^2 + 2\theta^2 + \frac{\sigma^2 d(d+1)}{4}.$$

The n conditionally independent centered matrices $W_i - E$ have zero expected pairwise Frobenius inner products. Their average therefore has exactly $1/n$ times this variance. Since $C - A_* = n^{-1} \sum_i (W_i - E)$, both claims follow. $\square$

Corollary 3.5 (Contraction with genuinely fresh blocks). *Initialize a deterministic finite B_0 and repeat the unprojected update in Theorem 3.4, with block size n and fresh independent blocks. Set*

$$\kappa = \frac{c_d + 2d}{n}, \quad \nu = \frac{\sigma^2 d(d+1)}{4n}.$$

For all integers $T \geq 0$,

$$\mathbb{E}\|B_T - A_*\|_F^2 \leq \kappa^T \|B_0 - A_*\|_F^2 + \nu \sum_{j=0}^{T-1} \kappa^j. \quad (11)$$

If $n > c_d + 2d$, the sum equals $(1 - \kappa^T)/(1 - \kappa)$ and $\kappa < 1$. If instead every update is replaced by a best symmetric rank-at-most- r Frobenius approximation, with known $r \geq \text{rank}(A_)$, the same statement holds with κ replaced by 4κ and ν by 4ν . That sufficient contraction condition is $n > 4(c_d + 2d)$.*

Proof. Cauchy–Schwarz on the eigenvalues gives $\text{tr}(E)^2 \leq d\|E\|_F^2$. Apply this inequality in (9), then take expectations. The resulting recurrence is $e_{t+1} \leq \kappa e_t + \nu$, where $e_t = \mathbb{E}\|B_t - A_*\|_F^2$. The right side is finite by induction; iterating the recurrence proves (11). For rank truncation, the feasibility of A_* and best-approximation property give

$$\|P_r(C) - A_*\|_F \leq \|P_r(C) - C\|_F + \|C - A_*\|_F \leq 2\|C - A_*\|_F.$$

Square and condition on the past to obtain the stated factors of four. A fixed measurable tie convention suffices to define the projection. $\square$

This result has an $O(d^2)$ sufficient block size and an unconstrained output that can require d signed neurons. A supplied rank is privileged information. These facts do not establish an optimal $O(dr)$ sample bound, selected width, or a computational advantage over fixed-width training. The matrix update is stochastic gradient descent for the matrix-coordinate loss in (6); it is not the same update as gradient descent in neuron factors. The exact variance formula also quantifies the familiar control-variate effect of subtracting a previously fitted predictor before estimating a residual.

3.4 A wrong surrogate can propose harmful growth at an exact fit

Theorem 3.6 (T4: false growth at the optimum). *Assume $\sigma = 0$ and $A_* \neq 0$. From any $n \geq 1$ fresh observations form S_n in (5), and consider the exact fit $B = A_*$. Put $D = S_n - A_*$ and let (λ, v) be a top-absolute-value unit eigenpair of D . Consider the signed neuron update $B' = A_* + \lambda v v^\top$. Its true risk increase and its improvement in the surrogate $\tilde{R}_{S_n}(B) = 2\|B - S_n\|_F^2$ coincide:*

$$R(B') - R(A_*) = \tilde{R}_{S_n}(A_*) - \tilde{R}_{S_n}(B') = 2\lambda^2. \quad (12)$$

This quantity is strictly positive almost surely. With $V = c_d\|A_\|_F^2 + 2\text{tr}(A_*)^2$,*

$$\frac{2V}{dn} \leq \mathbb{E}[R(B') - R(A_*)] \leq \frac{2V}{n}. \quad (13)$$

In contrast, the actual empirical residual moment at A_ is exactly zero on every sample.*

Proof. Since $Y_i = q_{A_*}(X_i)$, every actual residual at A_* is zero, proving the last assertion. The population risk identity and $\|vv^\top\|_F = 1$ yield $R(B') - R(A_*) = 2\lambda^2$. Also $\langle D, vv^\top \rangle = v^\top D v = \lambda$, so

$$\|D - \lambda v v^\top\|_F^2 = \|D\|_F^2 - 2\lambda^2 + \lambda^2 = \|D\|_F^2 - \lambda^2.$$

This proves the surrogate identity.

Apply Theorem 3.4 with $B = 0$ and $\sigma = 0$ to obtain $\mathbb{E}\|D\|_F^2 = V/n > 0$. At least one entry of D therefore has positive second moment. Every entry is a polynomial in the nd independent Gaussian coordinates. At least one is a nonzero polynomial; its zero set has Lebesgue measure zero. For completeness, this property follows by induction on the number of variables: view a nonzero multivariate polynomial as a polynomial in its last variable; outside the common zero set of its nonzero coefficient polynomials, it has finitely many roots, and Fubini’s theorem finishes the induction. The joint Gaussian law has a density, so $\mathbb{P}(D = 0) = 0$. Thus $|\lambda| = \|D\|_{\text{op}} > 0$ almost surely. Finally, $\|D\|_F^2/d \leq \|D\|_{\text{op}}^2 \leq \|D\|_F^2$; expectations give (13). $\square$

The exact fit is an oracle diagnostic checkpoint, not a checkpoint claimed to be learned by the experimental controller. The theorem concerns the unrestricted positive-score surrogate rule, with signed additions and without a nonzero uncertainty threshold. It does not assert that a thresholded or independently audited rule must accept. Any such accepted nonzero update at the exact fit is harmful, but its acceptance probability and expected accepted loss require a separate analysis. The theorem is not a lower bound for every algorithm using S_n .

3.5 A conditional width and risk certificate

Theorem 3.7 (Hard-threshold oracle inequality). *Let A_*, S be arbitrary symmetric matrices and let $b \geq 0$ satisfy $\|S - A_*\|_{\text{op}} \leq b$. Let $s_1(A_*) \geq \dots \geq s_d(A_*)$ denote the singular values of A_* , and let $\hat{A}$ retain exactly the eigenvectors of S with absolute eigenvalue strictly greater than $2b$. Then*

$$\text{rank}(\hat{A}) \leq \#\{j : s_j(A_*) > b\}, \quad (14)$$

$$\|\hat{A} - A_*\|_F^2 \leq \min_{0 \leq r \leq d} \left\{ 3 \sum_{j>r} s_j(A_*)^2 + 16b^2 r \right\}. \quad (15)$$

The corresponding excess prediction risk under (1) is twice the bound in (15).

Proof. The singular-value perturbation bound $|s_j(S) - s_j(A_*)| \leq \|S - A_*\|_{\text{op}}$ implies that $s_j(S) > 2b$ entails $s_j(A_*) > b$, proving (14). One way to see this perturbation bound is to apply the variational characterization $s_j(M) = \min_{\dim W = d-j+1} \max_{u \in W, \|u\|=1} \|Mu\|$ and the triangle inequality, then interchange S and A_* .

Hard thresholding at $2b$ minimizes, over symmetric B ,

$$\|S - B\|_F^2 + 4b^2 \text{rank}(B).$$

To justify this without a separate approximation theorem, let B have rank at most r and let P be the orthogonal projector onto its column space. Then

$$\|S - B\|_F^2 \geq \|(I - P)S\|_F^2 = \text{tr}(S^2) - \text{tr}(PS^2).$$

In an eigenbasis of S^2 ordered by decreasing eigenvalue, write $p_j = u_j^\top P u_j$. Since $0 \leq p_j \leq 1$ and $\sum_j p_j = \text{rank}(P) \leq r$, $\text{tr}(PS^2) \leq \sum_{j=1}^r s_j(S)^2$. Retaining the largest r singular components attains this bound and is realizable as a symmetric eigenvector sum. For each component, retaining its value costs $4b^2$, whereas omitting it costs its squared eigenvalue. At equality both choices minimize the objective, and the theorem specifies omission.

Fix $r \in \{0, \dots, d\}$ and let $A_{*,r}$ be a best symmetric rank-at-most- r approximation to A_* . Put $k = \text{rank}(\hat{A})$, $e = \|\hat{A} - A_*\|_F$, and $a = \|A_{*,r} - A_*\|_F$. Comparing penalized objectives and expanding around A_* gives

$$e^2 \leq a^2 + 2\langle S - A_*, \hat{A} - A_{*,r} \rangle + 4b^2(r - k).$$

The use of r rather than the possibly smaller rank of $A_{*,r}$ can only enlarge the right side. Operator/nuclear duality, the inequality $\|M\|_* \leq \sqrt{\text{rank}(M)}\|M\|_F$, and the triangle inequality imply

$$e^2 \leq a^2 + 2b\sqrt{k+r}(e+a) + 4b^2(r-k).$$

Apply $2b\sqrt{k+r}e \leq e^2/2 + 2b^2(k+r)$ and $2b\sqrt{k+r}a \leq a^2/2 + 2b^2(k+r)$. Rearranging yields $e^2 \leq 3a^2 + 16b^2r$. Finally, $a^2 = \sum_{j>r} s_j(A_*)^2$, and r was arbitrary. The prediction statement follows from Theorem 3.1. These arguments also cover $b = 0$; in that case $S = A_*$ and $\hat{A} = A_*$ directly. $\square$

The strict inequality in the threshold is necessary for the stated width bound: for $d = 1$, $b = 1$, $A_* = 1$, and $S = 2$, retaining the component at equality gives rank one, while the right side of (14) is zero. The risk bound alone allows either choice at equality because both minimize the penalized objective.

An explicit, conservative finite-sample radius follows without assuming an optimal concentration theorem. For the raw moment S_n , let

$$V_* = c_d \|A_*\|_F^2 + 2 \text{tr}(A_*)^2 + \frac{\sigma^2 d(d+1)}{4}.$$

Estimator	Noise SD 0	Noise SD .1
Pooled raw moment	0.037405	0.037583
Fresh residual correction	4.33×10^{-7}	0.000826
Known-rank correction	8.73×10^{-9}	0.000306
Lifted least squares	2.73×10^{-31}	0.000092

Table 2: Mean excess squared prediction risk after 4,096 labels over 64 independent seeds. These are estimator diagnostics, not trained-factor neural trajectories. Full intervals and raw outcomes are in `results/moments/findings.json`. The last two rows have explicitly different prior information or retained-state requirements.

Theorem 3.4 gives $\mathbb{E}\|S_n - A_*\|_F^2 = V_*/n$. For any deterministic known $\bar{V} \geq V_*$ and $\delta \in (0, 1)$, take $b = \sqrt{\bar{V}}/(n\delta)$. Markov’s inequality and $\|M\|_{\text{op}} \leq \|M\|_F$ show that the assumptions of Theorem 3.7 hold with probability at least $1 - \delta$. If $\bar{V} = 0$, the moment is exact almost surely. Unknown teacher norms cannot simply be substituted by same-sample point estimates without additional uncertainty control. This radius is conservative and signal dependent; it does not establish the optimal noisy low-rank rate. The oracle inequality is standard hard-threshold reasoning and is included to make that limitation explicit.

4 Reproducible diagnostics of the exact mechanism

The estimator study was specified in `research/MOMENT_DIAGNOSTIC_PROTOCOL.md` before execution. It has 64 paired independent seeds, two noise levels, dimension eight, and eight fresh batches of 512 observations. For each seed the unknown teacher is a random rotation of $\text{diag}(1, .3, 0, \dots, 0)$. The four estimators see the same chronological observations: pooled raw moment; fresh-residual full-matrix correction; rank-two-truncated correction; and lifted least squares. The rank-two arm is explicitly privileged by knowing the teacher rank. Lifted least squares retains the normal-equation Gram and right-hand side in 36-dimensional symmetric coordinates. It is a competent larger-state control, not an equal-storage comparator.

The raw archive contains all 128 input/label arrays and every stage’s four matrix estimates, yielding 1,024 stage records. An independent reconstruction recomputed all four estimators, risks, moment expectations and false-birth increases; maximum estimate discrepancy was 1.78×10^{-15} and maximum risk discrepancy 3.56×10^{-15} . Separate five-point tensor Gauss–Hermite integration verifies the polynomial identities for 20 matrices in dimensions one through four. At five nodes per coordinate it integrates these degree-eight polynomials exactly in ideal arithmetic; floating-point agreement is a check rather than the proof. Tied signs, all-zero spectra, strict-threshold equality and the invalid same-block conditioning example were also checked.

The noiseless fresh correction removes much of the raw estimator’s signal-induced error. This supports the mechanism, but not a novel statistical advantage: ordinary lifted least squares solves this small well-specified regression essentially to machine precision. Under noise it also has lower risk. Beating the raw moment alone therefore would be a weak algorithmic comparison.

At the exact noiseless teacher, every tested stale-summary birth increases true risk; the smallest recorded increase is .004771. This evaluates the oracle checkpoint in the negative theorem, not an implemented learner secretly receiving the teacher. Its purpose is to falsify the claim that $S_n - B$ equals finite-sample residual correlation. A valid uncertainty gate could reject these proposals. Safe rejection, however, would not prove that a useful next feature had been discovered.

The exact expected final raw risk is .041567 without label noise, whereas the 64-run mean is .037405 with descriptive t interval [.034092, .040718]. The expectation lies outside this particular interval. We preserve that discrepancy: the statistic has a heavy-tailed polynomial Gaussian

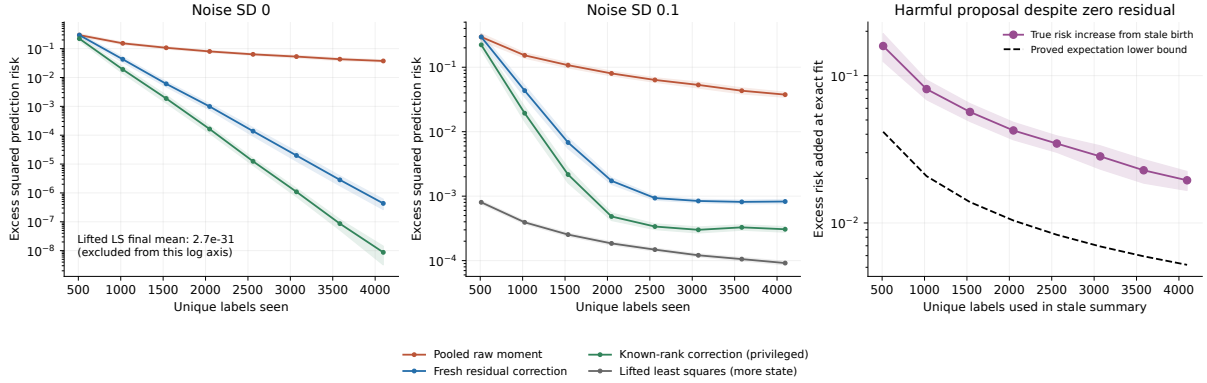


Figure 1: Saved estimator diagnostics. Shading is a descriptive 95% t interval across 64 runs. Left and center compare chronological estimators; the essentially zero noiseless lifted-LS endpoint is annotated instead of stretching the log axis. Right evaluates the *privileged exact-fit* checkpoint and compares the harmful stale-birth increase with its proved expectation lower bound. It is not a trained-method trajectory.

distribution, and a nominal finite-replicate t interval is not an exact confidence procedure for its mean. Independent exact quadrature and raw reconstruction support the identity; simulations neither replace its proof nor guarantee nominal interval coverage. The analogous noisy mean uses paired inputs and is not an independent replication of this fluctuation. No additional seeds were selected to improve the agreement.

The conditional width theorem is also not automatically a useful finite-sample selector. Here $V = 85.13$ without noise. At $n = 4096$ and $\delta = .05$, the crude valid radius is $b = \sqrt{V/(n\delta)} \approx .645$; its threshold $2b$ exceeds the largest teacher eigenvalue one. A conservative bound can reject nearly all growth despite an easy learnable problem. This separates an existence/validity calculation from useful selection power.

The simulator retains the entire generated archive for auditing, including all arms and teacher matrices. Named array sizes are not total persistent or peak memory measurements. An independent reviewer caught an ambiguous metadata field; the originally executed source and metadata are preserved, and `ACCOUNTING_CORRECTION.md` states the correction. The numerical code and outputs did not change. No memory superiority is inferred from this diagnostic. The separate neural studies count learner arrays and sample accesses under their own explicit interfaces.

5 A controlled learner with discovered quadratic directions

We next ask whether the distinction between raw target moments and residual moments improves an actual trained learner. This study implements the mechanisms as conventional ingredients; it is not a new spectral algorithm. No candidate receives a teacher direction. The principal negative result is that a better initial direction need not retain its advantage after fully trainable refitting, and an apparent final-risk benefit is sensitive to a simple replay control.

5.1 Frozen populations, data access and optimization

All inputs have distribution $N(0, I_{12})$. Two quadratic conditional means are $q_A(x) = x^\top Ax - \text{tr}(A)$, with nonzero eigenvalues $(.8, .45)$ and $(.65, -.5, .3, -.2)$, respectively. The first has no observation noise; the second has independent Gaussian noise with standard deviation $.3$. A null task has mean zero and noise standard deviation $.5$. The nonlinear task uses a hidden orthogonal rotation $z = U^\top x$ and mean $.8 \tanh(1.5z_1) \tanh(1.5z_2) + .4 \sin z_3$, with noise standard deviation $.2$. Task

rotations are fixed independently of run seeds and hidden from all learners. All labels, teacher parameters and evaluator inputs are saved.

The learned model is

$$f_{W,a,c}(x) = c + \sum_{j=1}^k a_j \{(w_j^\top x)^2 - \|w_j\|^2\}.$$

Every hidden vector, output coefficient and intercept trains. Initial vectors are normalized Gaussian draws and initial coefficients have standard deviation .01. Seven arrivals each reveal 1,024 labels. The current block and an old reservoir form the fitting pool; the block enters the reservoir only after fitting. Thus all 7,168 labels arrive at the same times, with no future data granted to the fixed comparator. Growing arms start at width one and receive six scheduled additions, ending at seven. This design isolates discovered directions; it does not claim autonomous width selection or safe rejection. New coefficients start at zero, preserving the current prediction before ordinary, potentially risk-increasing training.

Seven arms compare: historical-moment eigenbirth, old-reservoir residual eigenbirth, fresh-block residual eigenbirth, four learned residual probes, random growth with reservoirs of 128 and 138 examples, and a fixed width-seven online learner with 138-example replay. The historical arm stores 128 examples and $S_n = n^{-1} \sum_i y_i (x_i x_i^\top - I)/2$, and selects the eigenvector of largest absolute eigenvalue of $S_n - B$, where $B = \sum_j a_j w_j w_j^\top$. Residual eigenbirth instead forms $\{(\overline{y-f})xx^\top - (\overline{y-f})I\}/2$ on its permitted old reservoir or current block. It therefore includes the current intercept term. The zero-intercept theoretical variance formulas are not claimed as exact variance identities for every fitted checkpoint.

Learned probes maximize squared empirical residual covariance divided by candidate second moment plus .001, with twenty steps and four random starts. This is an established residual-feature objective. Directions are selected from observations and then all network parameters are refitted. Random additions use the same type of ordinary initial direction. The fixed learner knows the prescribed capacity cap, not the teacher rank.

Each arrival permits two million arithmetic-proxy units, including moment accumulation, candidate passes, eigendecomposition and fitting. Saved search work becomes additional training. Full-gradient work is charged at $3dk + 4k + 3$ per example; matrix and probe calculations have separate explicit charges in the runner. This is not matched hardware FLOPs. Persistent Adam uses learning rate .015, minibatches of at most 128, coefficients (.9, .999), denominator constant 10^{-8} , and gradient clipping at norm 20. New-unit optimizer moments are zero; existing optimizer state persists. No hyperparameter was changed using development evaluator outcomes.

Four development seeds on two tasks checked execution. The source, confirmation configuration and protocol were frozen at 05:40:48 UTC on 20 September 2026, before seeds 820000–820011. The resulting 48 paired configurations comprise 336 trajectories and 2,352 primary events, completed in 21.47 seconds using two CPU workers. Quadratic-task risks are evaluated exactly as $2\|B - A\|_F^2 + c^2 + \sigma^2$; nonlinear-task risk uses 32,768 independent evaluator inputs per seed and integrates label noise. Every trajectory is finalized before its evaluation. All reported intervals are descriptive 95% Student- t intervals across twelve independent seeds, without multiplicity adjustment.

5.2 Final risks and a failed attribution

On the indefinite task, historical moments appear favorable against fresh residuals: their paired final-risk difference is $-.005366$, with interval $[-.008879, -.001853]$. Their difference from the 138-example random-growth arm is $-.005141$ $[-.008227, -.002055]$. Those comparisons alone would support an overly strong interpretation. The 128-example random-growth arm also beats the 138-example arm by $-.005038$ $[-.007834, -.002243]$, while its mean risk is .097090, almost the same as historical moments at .096987. Their exploratory paired historical-minus-random-128

Method	Null	Rank two	Indefinite	Nonlinear
Historical moment	0.260725	0.000017	0.096987	0.161909
Replay residual	0.258093	0.000013	0.100358	0.164914
Fresh residual	0.258299	0.000044	0.102353	0.165770
Learned probe	0.256781	0.000087	0.106314	0.166361
Random R128	0.258551	0.000017	0.097090	0.160259
Random R138	0.257820	0.000021	0.102128	0.162947
Fixed width 7	0.256951	0.000014	0.116278	0.162604

Table 3: Mean final population MSE, generated from the frozen raw results. All methods finish at width seven. The rank-two task is noiseless; the other quadratic noise floors are .25 and .09. Nonlinear risks here use the prespecified Monte Carlo evaluator. Full intervals and every paired contrast are supplied as CSV.

difference is $-.000103$ $[-.001961, .001756]$. The near memory-matched contrast therefore does not isolate useful historical information. Independent code review found a concrete batching confound: the 128-example reservoir creates a 1,152-row fitting pool, divisible by the 128-row minibatch size; the 138-example reservoir creates 1,162 rows and an additional ten-row tail batch every epoch. The two random arms therefore perform 975 versus 1,034 Adam updates despite the same 124,285 fit-example accesses. Changes in update count, gradient noise and replay sample accompany the storage change. These results must not be interpreted as evidence that more memory harms learning. A separately frozen full-minibatch follow-up below tests this identified confound.

On the noiseless rank-two task, every method reaches a small mean error, ranging from 1.25×10^{-5} to 8.73×10^{-5} . The historical comparison with random 138 has interval $[-2.335, 1.608] \times 10^{-5}$; no final advantage is resolved. On the null task, all methods retain excess error above .25, illustrating that the scheduled-width study does not itself prevent needless capacity or optimization error. On the nonlinear task, historical moments versus random 138 has difference $-.001038$ $[-.006336, .004260]$, again unresolved. Intervals containing zero are not equivalence statements.

The fixed width-seven learner has competitive mean risk on three tasks and a worse, variable mean on the indefinite task. The historical-minus-fixed indefinite interval, $[-.039426, .000845]$, includes zero. This is a competent online baseline under the specified optimizer and budget, not proof that fixed models are optimal or that growing models uniformly dominate them.

5.3 Same-checkpoint directions and finite-time refitting

Lifetime histories diverge, so their final comparison cannot diagnose the quality of identical candidate sets. We therefore froze a separate, evaluator-only diagnostic. At every pre-growth checkpoint of the random-138 trajectory, all five direction mechanisms see the same current predictor, optimizer state, retained rows and available data prefix. Each direction is then followed by identical fitting work and minibatch randomness. Candidate work is additional and logged, rather than claimed equal. Reconstructing a historical moment from the archive explicitly charges all prefix rereads; it does not purchase those labels again or retroactively give that storage to the primary random learner. This adds 1,440 trained diagnostic branches.

For a quadratic task, the best possible output coefficient along a unit vector w yields gain $2\{w^\top(A - B)w\}^2$. This is computed from teacher parameters only after training and never enters selection. Averaged first within each seed over six checkpoints, historical directions exceed random directions in this potential gain by .08041 $[-.04305, .11777]$ on rank-two data and .13116 $[-.11530, .14702]$ on indefinite data. These are strong initial direction differences.

After identical fully trainable refitting, the corresponding historical-minus-random risk differences are 2.713×10^{-5} , with interval $[-1.111, 6.536] \times 10^{-5}$, and $-.002687$ $[-.006656, .001282]$,

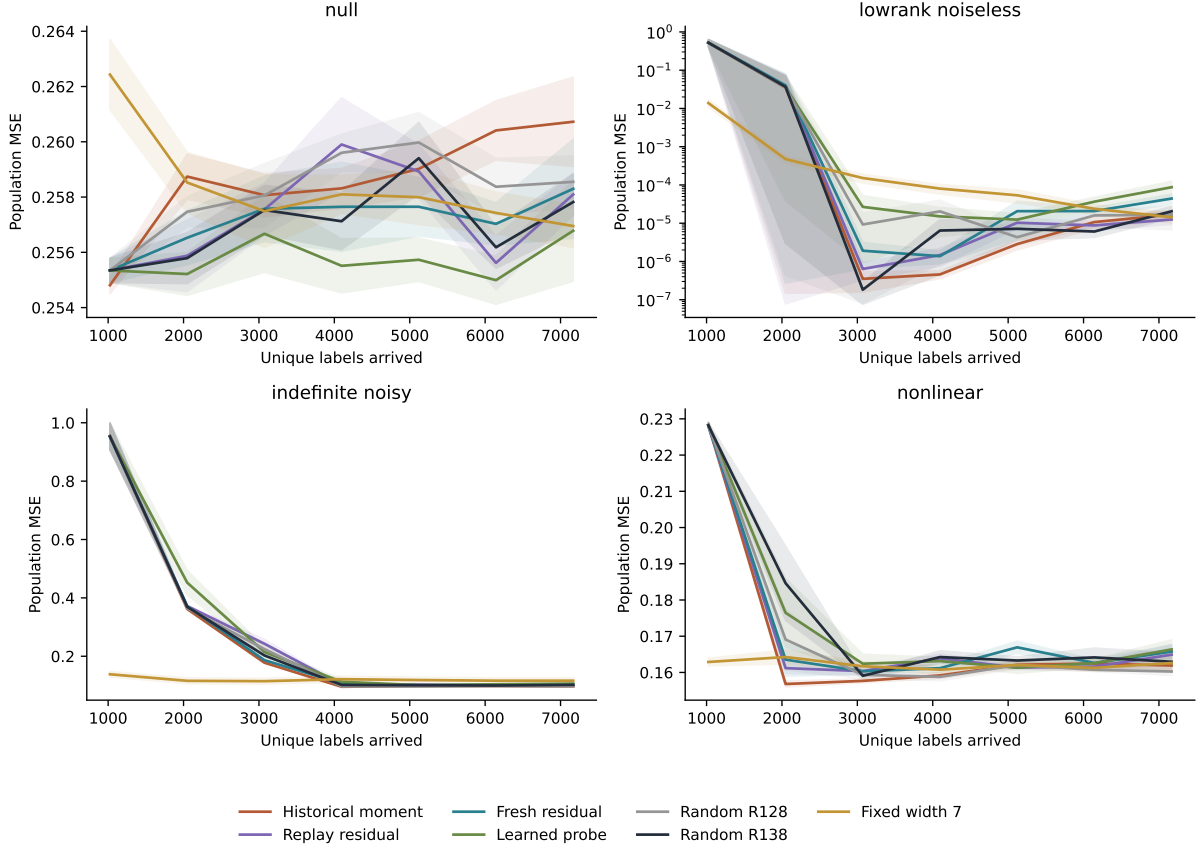


Figure 2: Complete trained histories from saved outcomes. Bands are one standard error across independent runs. All growing arms have the same scheduled widths; improvements here concern direction and fitting mechanisms. The rank-two panel uses a logarithmic risk axis.

respectively. Neither interval resolves a refitted advantage. Better direction alignment therefore does not by itself establish better learned models at the available training budget. This directly limits the proposed mechanism, rather than hiding a negative result behind a successful population direction formula. The trained hidden vectors may move substantially; this diagnostic does not claim that every source of erasure has been isolated.

5.4 Memory, computation and scientific limits

Every primary arm spends approximately fourteen million arithmetic-proxy units over its lifetime. Raw label accesses for fitting, proposals and moments are saved by sample ID. Historical retention uses $128(12+2)+144 = 1936$ eight-byte-word equivalents, including example labels and identities; random 138 uses 1932. At final width seven, both additionally use 276 model/Adam words and six controller-state words. This comparison is nearly matched in persistent algorithmic storage, not peak process memory. Current blocks, duplicated fitting pools, candidate matrices, hidden optimizer state and branch workspaces are separately reported. Python, BLAS/eigensolver internals and evaluator/archive storage are excluded from the array ledger.

Numerical differentiation checks the model gradient to maximum absolute error 3.52×10^{-10} and a zero-output birth preserves predictions to 3.47×10^{-17} . Raw archives contain observations, teacher/evaluator information in separate fields, all parameters and optimizer snapshots, source/configuration hashes and resource logs. Full result reconstruction and external novelty review are separate from these implementation checks. The main experiment neither establishes a new theorem nor a general neural advantage. Its contribution to the investigation

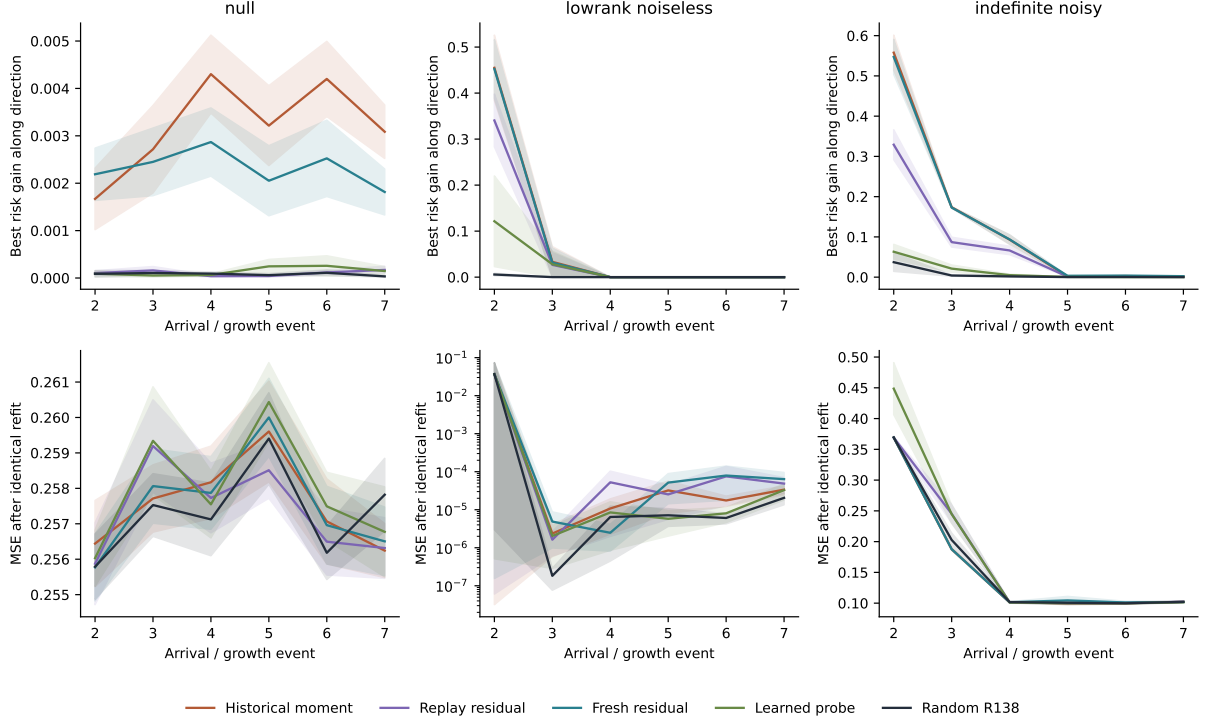


Figure 3: Same-checkpoint diagnostics. Top: evaluator-only best gain along the supplied-to-refitting, data-discovered direction. Bottom: actual risk after identical nonlinear refitting. These are different objects, and the large initial direction gap need not remain a final learning advantage.

is a reproducible attempt to learn structural directions, a failed attribution control, and direct separation of population opportunity from finite-sample nonlinear learning.

6 A separate attempt at adaptive width

The controlled-direction study fixes width growth. To make a serious attempt at actual structural adaptation, we separately implemented and froze a fresh-seed extension that can grow, continue fitting, or hold. This extension was designed after the main study and is exploratory; its outcomes do not retroactively confirm a preselected main-study algorithmic claim.

6.1 A deployable but heuristic controller

The task laws, model, seven 1,024-label arrivals, arithmetic budget and optimizer remain as above. Each arm retains 138 examples. Split each incoming block into 768 fitting/proposal labels and 256 validation labels. All candidate parameters are finalized before the current validation labels are inspected. After the action is committed, release the entire block into the reservoir, so completed validation labels can support future fitting. Warmup on the initial 768 labels is ungated. Future arrivals remain inaccessible.

Five arms compare fresh-residual growth, random growth, fixed width two with validation, fixed width seven with validation, and fixed width seven without validation. Growing learners begin at width one and can reach seven. At each later opportunity, fit both a continuation and a zero-output-unit growth branch from cloned incumbent optimizer states, splitting the remaining event budget equally after charging proposals and validation predictions. Fixed arms spend their remaining budget on continuation. Every rejected branch is charged.

For a candidate adding $r \in \{0, 1\}$ units, let $\bar{D}$ and $\widehat{\text{se}}(D)$ be the mean and sample standard

Method	Null	Rank two	Indefinite	Nonlinear
Fresh residual growth	0.25366 (1.08)	0.00014 (2.00)	0.10033 (4.08)	0.16204 (2.00)
Random growth	0.25501 (1.17)	0.00018 (2.08)	0.10499 (4.17)	0.16220 (2.00)
Fixed 2 gated	0.25763 (2.00)	0.07105 (2.00)	0.52321 (2.00)	0.18419 (2.00)
Fixed 7 gated	0.26259 (7.00)	0.00052 (7.00)	0.10260 (7.00)	0.16316 (7.00)
Fixed 7 ungated	0.26506 (7.00)	0.00002 (7.00)	0.10175 (7.00)	0.17007 (7.00)

Table 4: Separate adaptive extension: mean final MSE, followed by mean committed width in parentheses. All results use new seeds; they are not paired with the main study’s seeds. Full marginal and paired intervals are saved.

error of old-minus-new squared losses on the 256 validation points. Qualify it when

$$\bar{D} > 2\hat{\text{se}}(D) + .002 + .001r.$$

Install the qualifying candidate with largest mean gain minus $.002 + .001r$, or hold the exact incumbent if neither qualifies. This familiar holdout pattern and the standard-error heuristic are not novel. Unbounded losses, finite samples and unadjusted selection mean that we claim no population safety guarantee. A rejection does not certify that the current representation is adequate. This is deliberately separated from the earlier project’s valid bounded-loss audit theorem.

The protocol and source were frozen at 05:43:47 UTC on 20 September 2026, before seeds 840000–840011. The 48 new paired configurations contain 240 complete trajectories and 1,680 events; all completed in 13.38 seconds on two CPU workers. Source hashes, validation statistics, rejected branch parameters, optimizer states, retained sample identities and all data access counts are saved. A preflight perturbation of current validation labels leaves current branch parameters unchanged and leaves earlier committed histories unchanged. Descriptive intervals again use the twelve independent seeds, without a multiple-comparison correction.

6.2 Actual adaptive outcomes and retained failures

Fresh-residual growth learns mean final widths 1.08, 2.00, 4.08 and 2.00 on null, rank-two, indefinite and nonlinear tasks. The respective accepted additions across twelve runs are 1, 12, 37 and 12. On the indefinite task, the method therefore makes repeated learned changes within one lifetime, rather than restarting from a specially chosen point. Random growth reaches widths 1.17, 2.08, 4.17 and 2.00, with 2, 13, 38 and 12 accepted additions. All intermediate holds and continuations are retained in the raw trajectories.

The fresh-minus-random final-risk difference on the indefinite task is $-.004664$, with interval $[-.008976, -.000352]$. This favorable descriptive result is limited to one task among several comparisons. Their differences on null, rank-two and nonlinear tasks have intervals containing zero. In particular, information-guided proposal learning has not earned a general claim to superiority over random structural additions.

The wide fixed baselines limit the adaptive claim further. On indefinite data, fresh-minus-fixed-seven gated is $-.002269$ $[-.005805, .001266]$ and fresh-minus-fixed-seven ungated is $-.001418$ $[-.006792, .003955]$: no resolved final-risk advantage, though adaptation retains fewer units. On rank-two data the ungated fixed-seven learner instead reaches mean error 2.15×10^{-5} versus 1.44×10^{-4} for fresh growth, with paired adaptive-minus-fixed interval $[-.00000771, .00023663]$. Thus conservative commitments can stop useful further fitting. The fixed-two gated arm is especially variable or underfits; it is not used as the sole comparator.

Null outcomes distinguish restraint from learning adequacy. Fresh growth holds at 67 of the 72 post-warmup opportunities and usually remains at width one, but its mean risk .253656 still exceeds the Bayes risk .25. Its rare accepted addition is not evidence that a true structural signal exists. Good rejection behavior alone does not remove warmup or optimization error.

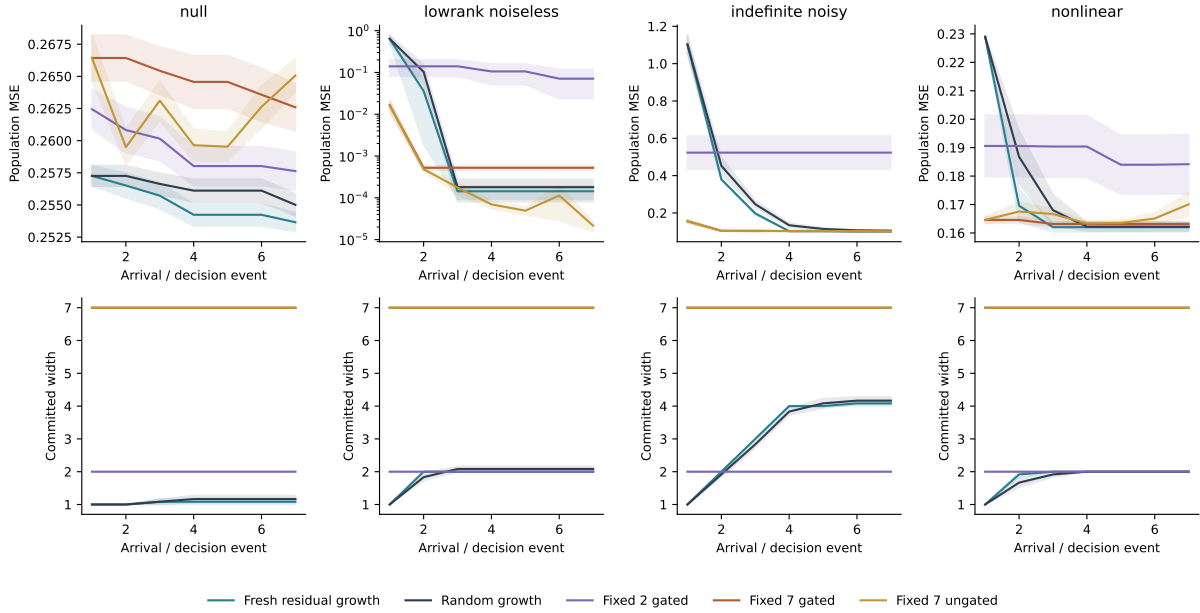


Figure 4: Separate fresh-seed adaptive histories. The upper row shows risk; the lower row shows actually committed widths. Bands are one standard error. The opportunity schedule is precommitted, while directions and accepted widths are chosen from observed data.

The nonlinear task similarly remains constrained by the chosen model family, even when the controller stops adding units.

No accepted fresh-growth or random-growth step increased the evaluator risk in these runs, but this observation is not a coverage guarantee. The same heuristic did accept one evaluator-risk increase in the fixed-two rank-two runs and one in its nonlinear runs; the latter evaluator is Monte Carlo. All are retained. Ungated fixed-seven training has 33, 8, 28 and 36 risk-increasing steps across the four tasks. These numbers describe sampled optimization histories and do not establish the statistical validity of the heuristic gate.

6.3 Cost and interpretation

All arms consume approximately fourteen million charged arithmetic-proxy units. For fresh growth, rejected branches alone consume mean totals of 10.62, 9.83, 7.79 and 9.83 million across the four tasks. These costs are not hidden behind the smaller deployed width. Fixed wide learners have greater persistent optimizer storage from the start, while adaptive candidates require additional simultaneous branch storage; the separate ledgers expose both. No equal-width or equal-proxy statement is promoted to equal physical memory or equal wall-clock computation.

This extension realizes data-dependent architecture selection with learned hidden vectors, ordinary starts, label release and strong online controls. It does not establish a novel growth principle, a valid universal gate, optimal rank discovery, or publication readiness. Its favorable indefinite comparison, negative fixed-capacity comparison, false-risk-decrease episodes, and substantial rejected computation are all part of the result.

7 A targeted response to the minibatch confound

We did not stop at documenting the different tail batches in the original replay comparison. A separately frozen follow-up clones the primary runner and changes the fitting batch assembly: carry the end of a shuffled pool into a new independent permutation until a full 128-example batch is formed. Only the final arithmetic-budget remainder can form a smaller batch. All models, task

Method	Null	Rank two	Indefinite	Nonlinear
Historical moment	0.260777	0.000022	0.096205	0.162987
Random R128	0.259258	0.000047	0.098319	0.161161
Random R138	0.259687	0.000012	0.096821	0.163098
Fixed width 7	0.259182	0.000021	0.104035	0.159968

Table 5: New-seed, full-minibatch follow-up. These are separate observations, not overwritten entries in Table 3. All means are final population MSE, with the same nonlinear Monte Carlo evaluation rule.

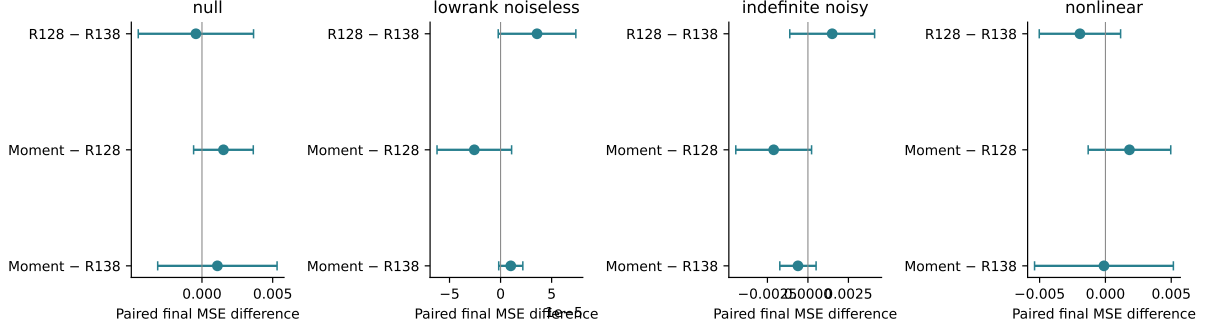


Figure 5: Targeted paired risk differences after removing epoch-tail changes in update count. Bars are descriptive 95% Student- t intervals over twelve independent new seeds; no multiplicity adjustment is made. None of these intervals establishes a resolved final-risk advantage.

laws, proposal rules, optimizer settings, data arrivals and resource charges remain unchanged. The original source and outcomes are preserved. This is a targeted post-hoc investigation, not a retroactive replacement of the main experiment.

Before running seeds 850000–850011, we froze the derived source, four-arm configuration and protocol at 05:49:36 UTC on 20 September 2026. The arms are historical moments with replay 128, random growth with replay 128 and 138, and fixed width seven with replay 138. We omit the separate paired-branch diagnostic here, as specified before execution. The 48 new configurations produce 192 complete trajectories and 1,344 events in 9.76 seconds. Every random-128 and random-138 run now has exactly 975 optimizer updates and 124,285 fitting-example accesses. Their example counts, update counts and minibatch-size sequences are thus matched, while their retained samples can still differ. Historical moments spend candidate work and use 892 updates; fixed width seven uses 392 updates, reflecting its larger per-example cost.

The original adverse indefinite-task comparison between replay sizes does not persist. Random-128 minus random-138 is now $+.001498$, with interval $[-.001121, .004117]$, compared with the original negative estimate $-.005038$ and interval $[-.007834, -.002243]$. Historical moments minus random-138 is $-.000616$, with interval $[-.001736, .000504]$; historical moments minus random-128 is $-.002114$, with interval $[-.004453, .000226]$. Both intervals include zero. Thus a general learning benefit from retaining the target moment is not established by this follow-up. On the other tasks, all targeted historical and replay-size paired intervals likewise include zero. That statement is a failure to resolve effects, not an equivalence proof.

The changed update schedule is mechanically verified, and the new empirical comparisons fail to support the original strong attribution. Because the follow-up uses new seeds, the difference between old and new point estimates is not itself a paired causal estimate of the minibatch correction. The correct conclusion is narrower: the original comparison was confounded, and the proposed moment advantage has not survived the present control with matched minibatch schedules. A stronger assessment would randomize or pair both batching schedules on an additional untouched seed set, ideally with an optimizer-budget definition chosen in advance.

Across the primary, adaptive and targeted studies there are 768 complete confirmation/follow-up trajectories, plus 1,440 same-checkpoint trained branches. This volume is evidence of implementation and investigation, not a substitute for a surviving theorem-level novelty claim or a robust empirical improvement. The unsuccessful attribution remains part of the research result.

8 What the investigation establishes and leaves unresolved

The distinction between a population inverse and empirical residual subtraction is exact and matters even without label noise. A stored matrix can contain genuinely data-discovered directions while repeated processing of it adds no new evidence. Updating that matrix by subtracting current predictions at the sample level is a different operation, and fresh inputs permit a clean conditional calculation. These are useful constraints on the original information-driven learning idea: the relevant information must support the *current* optimization operator, not only its expectation at a fixed independent model.

The current results do not finish the intended contribution. They provide no finite-precision lower bound for all learners, no optimal unknown-rank sample-and-runtime result, no global guarantee for hidden-weight Adam, and no evidence that neuron-local information measures beat strong existing search procedures. The Gaussian model is a controlled setting with an exact population inverse; unknown or non-Gaussian covariate laws require additional estimation. The matrix correction theorem is not the trained factor-Adam algorithm, and the main direction experiment has scheduled widths. A heuristic gate does not acquire a false-acceptance theorem by sharing a fresh-block data schedule.

Storage is accounted for through named arrays and access logs; physical peak memory and floating-point roundoff are not bounded theoretically. Inference width, persistent learner state, working arrays, saved scientific records, and evaluator storage are separate resources. Near-matched replay storage does not imply matched total memory, exact FLOPs, or hardware time. Uncertainty intervals use independent seeds, not time points or branch proposals. They are descriptive and not multiplicity-adjusted. The finite synthetic task set supports no universal performance claim.

A substantive next theorem would need a consequence beyond existing optimal recovery, coresets approximation, spectral thresholding, and stochastic low-rank optimization. One precise target is noisy, unknown-rank online discovery with an explicit approximation oracle bound and an accountable advantage under a declared memory-work frontier. A proof must not silently import an ordinary RIP that the measurements lack, and an empirical improvement must survive competent fixed-rank and matrix-free alternatives. The present evidence does not establish that such an advantage exists. If these comparisons subsume the next candidate, it should be rejected rather than renamed.

9 Reproduction and evidence integrity

The companion workspace preserves source versions, raw observations, per-run parameters, fitting/proposal accesses, all figure scripts, and independent reports. `README.md` gives a minimal verification command and complete re-execution instructions into new destinations. Each confirmatory protocol was frozen before its own confirmation seeds ran. Development fixes, source-hash differences, accounting corrections and post-hoc analytic diagnostics are recorded explicitly. Prior artifacts are checked against the baseline manifest. No earlier result is overwritten, and no external publication, submission, paid compute or private-data upload was performed.

A A literal-assumption caveat in a current spectral theorem

We checked Defilippis et al. [6], arXiv:2602.05846v1, on 20 September 2026 UTC: Definitions 2.1–2.7 and 3.3, Assumptions 2.3–3.4, Theorem 3.6, and Appendix A.2. The current record listed only v1, dated 5 February 2026. The spectral estimator uses

$$M = \sum_{i=1}^n \mathcal{T}(Y_i) X_i X_i^\top, \quad X_i \sim N(0, I_d/d),$$

with eigenvalue-gap selection and eigenvector norm $\sqrt{d}$. The preprocessing conditions are boundedness, positive essential supremum, and a nonzero value with positive probability. We test only the literal universal preprocessing scope of the rate claim, not other results or a qualified theorem. This is not our main originality claim.

Proposition A.1 (Constant preprocessing fails the literal universal claim). *Take orthogonal target directions w_k^* with $\|w_k^*\|_2^2 = d$, $1 \leq k \leq m_* \leq d$, and normalized weights*

$$a_k = \frac{k^{-\gamma}}{\left(\sum_{j=1}^{m_*} j^{-2\gamma}\right)^{1/2}}, \quad \gamma > \frac{1}{2}.$$

Let $g_k(z) = (z^2 - 1)/\sqrt{2}$ and

$$Y = \sum_{k=1}^{m_*} a_k g_k(\langle w_k^*, X \rangle) + \sqrt{\Delta} \xi, \quad \Delta > 0, \quad \xi \sim N(0, 1),$$

with X and ξ independent. Choose $\mathcal{T}(y) = 1$. For $n \geq d \geq 2$, select any number $r \leq m_*$ of the ordered eigenvectors of M by a rule depending only on its eigenvalues, normalize them to $\sqrt{d}$, and pad the remaining output rows by zero. The resulting weighted matrix reconstruction error satisfies

$$\sum_{k=1}^{m_*} \frac{a_k^2}{d^2} \mathbb{E} \left[\|\hat{w}_k \hat{w}_k^\top - w_k^* w_k^{*\top}\|_F^2 \right] \geq 1. \quad (16)$$

The expectation is over the input sample; the assertion also holds for random orthogonal teachers independent of those inputs.

Proof. The links are even polynomials with $\mathbb{E}[g_k(Z)^2] = 1$ and $\mathbb{E}[g_k''(Z)] = \sqrt{2}$ for $Z \sim N(0, 1)$. They have a uniform polynomial envelope and satisfy the displayed nondegeneracy conditions on the target. The weights are strictly decreasing, have sum of squares one, and satisfy the required power-law scaling with constants independent of m_* , because $\gamma > 1/2$. The preprocessing is bounded, is never zero, and has the specified positive essential supremum:

$$\inf\{c : \mathbb{P}(\mathcal{T}(Y) < c) = 1\} = \inf(1, \infty) = 1.$$

Thus it meets the literal preprocessing conditions.

With this choice, $M = \sum_i X_i X_i^\top$ depends only on isotropic Gaussian inputs. It is a rotationally invariant Wishart matrix, independent of any information about target directions in the labels. For $n \geq d$ its eigenvalues are distinct almost surely. Conditional on the ordered eigenvalues, every normalized rank-one eigenprojector is uniformly oriented. In particular, for an eigenvector v normalized by $\|v\|_2^2 = d$,

$$\mathbb{E}[vv^\top \mid \text{spec}(M)] = I_d, \quad \mathbb{E}[(v^\top w_k^*)^2 \mid \text{spec}(M)] = d.$$

These statements concern projectors and hence do not depend on how the sign of an eigenvector is chosen. Selection of r through eigenvalues also leaves them unchanged. Consequently every selected row has conditional normalized error

$$\frac{\mathbb{E}[\|vv^\top - w_k^* w_k^{*\top}\|_F^2 \mid \text{spec}(M)]}{d^2} = \frac{d^2 + d^2 - 2d}{d^2} = 2 - \frac{2}{d}.$$

A zero-padded row has normalized error one. If p_k is the probability that row k is selected, the left side of (16) is therefore

$$\sum_{k=1}^{m_*} a_k^2 \left\{ p_k \left(2 - \frac{2}{d} \right) + (1 - p_k) \right\} = 1 + \left(1 - \frac{2}{d} \right) \sum_{k=1}^{m_*} a_k^2 p_k \geq 1.$$

Conditioning first on a random teacher gives the same conclusion when the teacher is independent of the inputs. $\square$

The literal rate assertion combines Theorem 3.6 with the displayed large-sample regime $\alpha = n/d \gg m_*^{2\gamma}$, in which the claimed weighted optimal rate is of order m_*/α . That rate can tend to zero, whereas (16) cannot. For example, in the subsequent large- α, m_* comparison, taking $\alpha = m_*^{2\gamma+1}$ gives $m_*/\alpha = m_*^{-2\gamma}$. The disagreement persists after the proportional high-dimensional limit, since the lower bound holds at every finite $d \geq \max(m_*, 2)$. Likewise, normalized squared overlap of any selected row with a fixed target direction is $1/d$, so no fixed finite α yields weak recovery from this input-only estimator.

The source proof's transition from its equations (66) to (70) has a leading coefficient containing $\int \mathcal{T}(y)Z'(y)dy$. For the constant preprocessing this is zero: Z is a Gaussian convolution density, tends to zero at both ends, and has an integrable derivative, so its derivative integrates to zero. Boundedness and nonzero preprocessing do not imply a nonzero appropriately signed coupling to the target. An added coupling condition, or a specified valid preprocessing family, would be a substantive qualification to check in a corrected theorem. This observation does not supply a new learning mechanism for our project.

B Exact approximation floor for the nonlinear stress task

This analytic diagnostic was developed after the primary protocol was frozen. It does not change training, tuning, or the original Monte Carlo endpoint. Let Z_1, Z_2, Z_3 be independent standard normal coordinates after an orthogonal rotation, and let

$$f_*(Z) = .8 \tanh(1.5Z_1) \tanh(1.5Z_2) + .4 \sin Z_3.$$

Define

$$a = \mathbb{E}[Z \tanh(1.5Z)], \quad b = \mathbb{E}[\tanh^2(1.5Z)], \quad c = .8a^2.$$

Proposition B.1 (A width-independent approximation obstruction). *The orthogonal projection of f_* onto the class of all centered quadratic forms plus constants is cZ_1Z_2 . If u_1, u_2 are the corresponding orthonormal directions and $A_0 = \frac{c}{2}(u_1u_2^\top + u_2u_1^\top)$, then every quadratic network, of arbitrary width, with matrix B and intercept c_0 satisfies*

$$\mathbb{E}[(f_*(X) - q_B(X) - c_0)^2] = F_0 + 2\|B - A_0\|_F^2 + c_0^2,$$

where

$$F_0 = .64b^2 + .08(1 - e^{-2}) - .64a^4.$$

The best predictor is realizable by two signed quadratic neurons.

Proof. A centered quadratic form is a linear combination of $Z_i^2 - 1$ and $Z_iZ_j, i < j$. Independence and oddness imply that f_* has zero mean and zero inner product with every diagonal term. The product of two odd hyperbolic tangents has zero correlation with every cross term except Z_1Z_2 ; for that term the correlation is $.8a^2 = c$, while $\mathbb{E}[Z_1^2Z_2^2] = 1$. The sine term is odd under $Z \mapsto -Z$ and is orthogonal to every quadratic form and to constants. These facts identify the projection.

The two components of f_* are orthogonal, with squared norms $.64b^2$ and $.16\mathbb{E}\sin^2 Z = .08(1 - e^{-2})$, respectively; the latter follows from $\sin^2 z = (1 - \cos 2z)/2$ and the Gaussian characteristic function. Subtracting the projection's squared norm c^2 yields F_0 . The Pythagorean identity and Theorem 3.1 give the risk formula. Finally, A_0 has eigenvectors $(u_1 + u_2)/\sqrt{2}$ and $(u_1 - u_2)/\sqrt{2}$, eigenvalues $c/2$ and $-c/2$, so two signed neurons suffice. $\square$

One-dimensional numerical quadrature gives $a \approx .689027429$, $b \approx .540648381$, $c \approx .379807038$ and $F_0 \approx .111992221$. The added observation-noise variance .04 makes the best total squared risk approximately .151992221. These values are numerical evaluations of the displayed expectations, not additional statistical estimates. The script `code/nonlinear_projection.py` reproduces the quadrature and evaluates saved final parameters using this identity. Those post-hoc analytic evaluations are stored separately; all original Monte Carlo primary outcomes remain unchanged. The sine component alone contributes an irreducible .069173177 for this architecture. More width cannot remove this misspecification. Information-guided adaptation would need to change the function family, not merely discover additional quadratic directions.

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